



More Life

NASA astronaut, Kate Rubins, harvests radish crop grown on the International Space Station. <http://digit.in/jan21-19>



Ryugu Asteroid Dust

Japan's Hayabusa2 mission has brought back black sandy asteroid dust; reveals first image. <http://digit.in/jan21-20>

designating different bulge to disc ratios. For the spiral galaxies, the classification changes based on how tightly wound the arms are around the center. Irregular galaxies as you can imagine come in all kinds of amorphous blobs. The entire scheme is also known as the Hubble sequence, or the Hubble tuning fork.

Apart from the shape, the galaxies vary by luminosity and colour too. Some galaxies are a hundred times brighter than the Milky Way, and others are a thousand times dimmer. Then there are of course small and large galaxies, with the smaller ones being known as dwarfs, while the behemoths being known as giants. IC 1101, is a supergiant elliptical galaxy that stretches over 4 million light years across. Some galaxies spin, and other galaxies do not. In some spherical galaxies with low rotation, where the stars have thermal motions similar to water heated fluid molecules. Some galaxies are active with stellar nurseries and young stars, while other galaxies have old stars that have gone through many changes over their life spans. The galaxies also interact with each other, with the dwarf galaxies being swallowed up by the cannibal galaxies.

By just grouping the galaxies together, we can look at how they changed over time. The universe also has a built in time scale, because the further we see, the older the galaxies are. Some of the galaxies in the Hubble Deep Field for example, are the oldest and the farthest that we know of.

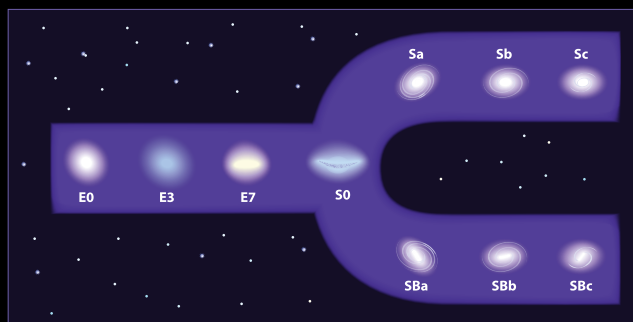
Soon after the Big Bang, the Universe was made up of mostly homogeneous matter, with only a few odd clumps of matter. These slight perturbations would go on to be the basis of the formation of the first accumulations of matter that gave rise to protogalactic clouds about 1 billion years after the birth of the cosmos. The first

stars, made up of only hydrogen and helium formed within these protogalactic clouds, and over their life spans they produced the heavier elements. The violent deaths of these first generation stars heated the galactic clouds, preventing them from collapsing back together under the influence of gravity, causing the gas to settle in a disc. The appearance of a galaxy depends on the type of stars that are formed, that is how massive they are and how long they burn. The high mass stars are short lived and bright, while the low mass stars are long lived but dim. The proportion of low mass

as they feed on others, becoming massive objects at the centers of galaxy clusters. Studies have shown that when two elliptical galaxies collide, the interactions cause the combined object to assume a spiral shape, in as little as a billion years. This goes to explain the observation that whenever there are more galaxies clustered together, the frequency of spiral galaxies towards the center of the cluster increases.

Most of the large galaxies, including the ellipticals and spirals that we see in the night sky have finished their evolution long ago. Over time, the larger galaxies that

cannibalise smaller ones, do not change much in their character, at least not as dramatically as two similar sized galaxies interacting. Astronomers are on the hunt for the



The Hubble "tuning fork"

and high mass stars determine the luminosity and colour of the galaxy. During periods of star formation, galaxies appear bluer. This is also true for the discs of the galaxies, which appear bluer because of increased star formation. The bulges of galaxies are generally redder because these are relatively static areas with little new star formation.

The less dense protogalactic clouds with less star formation formed the spiral galaxies. The more dense protogalactic clouds with more star formation formed the elliptical galaxies. As the distance between two galaxies is not much more than their diameters, it is inevitable that galaxies interact and collide over time. This leads to the formation of all kinds of shapes and sizes. Some galaxies grow larger and larger

so called early bright galaxies, the largest and brightest galaxies that the universe could have produced, about 13 billion years ago. These galaxies would radiate so much ultraviolet light, that they broke up the atoms in the surrounding space. These are expected to be unusually bright and large galaxies, with clouds of ions around them. While there have been candidate objects identified, including a galaxy known as CR7, it remains up to future instruments to investigate further, and humans are going to launch just such a telescope. While the Hubble Space Telescope let us look as far back as the formation of the first galaxies, the JWST promises to allow us to look at the formation of the first stars, which will allow scientists to get a much more complete picture of galaxy evolution. 